

LSH 00 — ENGINEERING DOCUMENT TREE

Sequential Controlled Document List (001 – 531)

Derived from LSH 00 Generation 1 | Revision 3.2 | Preliminary Controlled System Baseline

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This document defines the controlled engineering document tree for the Lightweight Safe Helicopter (LSH) Generation 1 concept with separable survival capsule. Document numbers run continuously from 001 to 531. Each entry retains its logical Document ID (for configuration management) and a Document Type. The tree respects LSH 00 Rev 3.2 engineering integrity rules: no unverified performance is treated as demonstrated.

Total documents: 531 (sequential numbers 001 – 531)

Major Branch	From	To	Count
1. SYSTEM LEVEL BASELINE & MANAGEMENT	001	026	26
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TOTAL	001	531	531

Columns: **No.** = continuous sequence | **Document ID** = controlled identifier | **Document Title** | **Document Type** = controlled vocabulary for CM.

1. SYSTEM LEVEL BASELINE & MANAGEMENT

Documents 001 – 026 (26 documents)

No.	Document ID	Document Title	Document Type
001	LSH-00	Lightweight Safe Helicopter — Comprehensive Engineering Baseline	System Baseline
002	LSH-SYS-001	System Requirements Specification (SRS)	Requirements
003	LSH-SYS-002	System Architecture Description	Architecture
004	LSH-SYS-003	System Concept of Operations (ConOps)	Concept
005	LSH-SYS-004	System Functional Breakdown Structure	Architecture
006	LSH-SYS-005	System Interface Control Document Master	ICD
007	LSH-SYS-006	Configuration Management Plan	Management Plan
008	LSH-SYS-007	Requirements Traceability Matrix Master	Traceability
009	LSH-SYS-008	System Risk & Opportunity Register	Risk Register
010	LSH-SYS-009	Open Items & Assumption Closure Plan	Management Plan
011	LSH-SYS-010	Master Verification Gates Definition (VG-LSH-01 to 12)	Verification Plan
012	LSH-SYS-011	System Mass Properties Accounting Model	Mass Properties
013	LSH-SYS-012	System CG & Inertia Envelope Definition	Mass Properties
014	LSH-SYS-013	System Design Constraints & Trade Study Log	Trade Study
015	LSH-SYS-014	Technology Readiness Assessment	Assessment
016	LSH-SYS-015	System Safety Philosophy & Layer Definition	Safety Philosophy
017	LSH-SYS-016	Document Tree & Data Management Plan	Management Plan
018	LSH-SYS-017	Engineering Change Control Procedure	Procedure
019	LSH-SYS-018	Master Glossary & Terminology Control	Reference
020	LSH-SYS-019	System Boundary & External Interface List	ICD
021	LSH-SYS-020	Preliminary Hazard List (System Level)	Safety Assessment
022	LSH-SYS-021	Investor Technical Due-Diligence Package Index	Index
023	LSH-SYS-022	IP Asset Register & Licensing Readiness	IP Package
024	LSH-SYS-023	Program Milestone & Review Gate Plan	Management Plan
025	LSH-SYS-024	System Performance Target Allocation Matrix	Requirements
026	LSH-SYS-025	Lessons Learned & Previous Revision Basis	Reference

2. SAFETY ASSESSMENT & CERTIFICATION

Documents 027 – 071 (45 documents)

No.	Document ID	Document Title	Document Type
027	LSH-SAF-001	Functional Hazard Assessment (FHA) — System	Safety Assessment
028	LSH-SAF-002	Preliminary System Safety Assessment (PSSA)	Safety Assessment
029	LSH-SAF-003	System Safety Assessment (SSA)	Safety Assessment
030	LSH-SAF-004	Common-Cause Analysis Report	Safety Assessment
031	LSH-SAF-005	Zonal Safety Analysis	Safety Assessment
032	LSH-SAF-006	Particular Risk Analysis	Safety Assessment

	Document ID	Document Title	Document Type
033	LSH-SAF-007	Fault Tree Analysis — Top Events	Safety Assessment
034	LSH-SAF-008	FMEA / FMECA — System Level	Safety Assessment
035	LSH-SAF-009	Unintended Jettison Quantitative Safety Objective	Safety Assessment
036	LSH-SAF-010	Fire & Fuel Hazard Analysis	Safety Assessment
037	LSH-SAF-011	Human Factors Safety Assessment	Safety Assessment
038	LSH-SAF-012	Safety Requirements Allocation Matrix	Requirements
039	LSH-SAF-013	Independence & Dependency Analysis of Safety Layers S1–S6	Safety Assessment
040	LSH-SAF-014	Catastrophic Failure Condition List	Safety Assessment
041	LSH-SAF-015	Safety Case Structure & Argument Map	Safety Case
042	LSH-SAF-016	Certification Basis Selection Report	Certification
043	LSH-SAF-017	Means of Compliance Matrix	Certification
044	LSH-SAF-018	Certification Plan	Certification
045	LSH-SAF-019	Type Design Definition Document	Certification
046	LSH-SAF-020	Conformity Plan	Certification
047	LSH-SAF-021	Continued Airworthiness Strategy	Certification
048	LSH-SAF-022	Special Conditions / Equivalent Safety Findings Log	Certification
049	LSH-SAF-023	Safety Review Board Charter	Management Plan
050	LSH-SAF-024	Hazard Tracking System Procedure	Procedure
051	LSH-SAF-025	Quantitative Risk Assessment Methodology	Methodology
052	LSH-SAF-026	Software Criticality & DAL Allocation	Safety Assessment
053	LSH-SAF-027	Complex Electronic Hardware Assurance Plan	Assurance Plan
054	LSH-SAF-028	Occupant Safety Criteria Definition	Requirements
055	LSH-SAF-029	Crashworthiness Requirements Allocation	Requirements
056	LSH-SAF-030	Emergency Egress & Post-Crash Survival Requirements	Requirements
057	LSH-SAF-031	Safety Assessment of Separation Sequence	Safety Assessment
058	LSH-SAF-032	Common Mode Analysis — Power & Sensing	Safety Assessment
059	LSH-SAF-033	Safety Assessment of Partial / Asymmetric Separation	Safety Assessment
060	LSH-SAF-034	Lightning & HIRF Safety Assessment	Safety Assessment
061	LSH-SAF-035	Icing & Environmental Hazard Assessment	Safety Assessment
062	LSH-SAF-036	Bird Strike & Foreign Object Impact Assessment	Safety Assessment
063	LSH-SAF-037	Safety Assessment of Ground Handling & Maintenance	Safety Assessment
064	LSH-SAF-038	Safety Data Package for Authority	Certification
065	LSH-SAF-039	Master Minimum Equipment List (M MEL) Philosophy	Certification
066	LSH-SAF-040	Operational Safety Assessment (OSA)	Safety Assessment
067	LSH-SAF-041	Safety Training & Awareness Requirements	Requirements
068	LSH-SAF-042	Incident / Anomaly Investigation Procedure	Procedure
069	LSH-SAF-043	Safety Performance Indicators Definition	Management Plan
070	LSH-SAF-044	Independent Safety Assessment Plan	Assurance Plan
071	LSH-SAF-045	Safety Closure & Evidence Index	Evidence Index

3. SURVIVAL CAPSULE SYSTEM

Documents 072 – 166 (95 documents)

	Document ID	Document Title	Document Type
072	LSH-CAP-001	Capsule System Requirements Specification	Requirements
073	LSH-CAP-002	Capsule Architecture Description	Architecture
074	LSH-CAP-003	Capsule Concept of Operations (Normal & Emergency)	Concept
075	LSH-CAP-004	Capsule Interface Control Document (Airframe)	ICD
076	LSH-CAP-005	Capsule Mass Properties Report	Mass Properties
077	LSH-CAP-006	Capsule CG & Inertia Model	Mass Properties
078	LSH-CAP-007	Capsule Primary Structure Design Description	Design Description
079	LSH-CAP-008	CFRP Roll-Cage Structural Analysis	Analysis Report
080	LSH-CAP-009	Metallic / Titanium Nodes Design Description	Design Description
081	LSH-CAP-010	Capsule Joint & Load-Path Analysis	Analysis Report
082	LSH-CAP-011	Capsule Damage Tolerance & Delamination Assessment	Analysis Report
083	LSH-CAP-012	Capsule Environmental Protection & Lightning Design	Design Description
084	LSH-CAP-013	Capsule Inspection & Repair Philosophy	Maintenance
085	LSH-CAP-014	Energy-Absorbing Seats Design Description	Design Description
086	LSH-CAP-015	Occupant Restraint System Requirements & Design	Requirements
087	LSH-CAP-016	Seat Dynamic Performance Analysis	Analysis Report
088	LSH-CAP-017	Landing Attenuation System Requirements	Requirements
089	LSH-CAP-018	Landing Attenuation Design Description	Design Description
090	LSH-CAP-019	Landing Attenuation Stroke & Load Analysis	Analysis Report
091	LSH-CAP-020	Failed-Leg & Asymmetric Landing Cases Analysis	Analysis Report
092	LSH-CAP-021	Capsule Floor & Load Distribution Analysis	Analysis Report
093	LSH-CAP-022	Capsule Internal Layout & Human Factors	Design Description
094	LSH-CAP-023	Capsule Door / Hatch Design & Emergency Egress	Design Description

Item	Document ID	Document Title	Document Type
095	LSH-CAP-024	Capsule Transparency & Visibility Requirements	Requirements
096	LSH-CAP-025	Capsule Thermal & Acoustic Environment	Analysis Report
097	LSH-CAP-026	Capsule Life-Support & Post-Event Systems	Design Description
098	LSH-CAP-027	Recovery Aids & Location Devices Design	Design Description
099	LSH-CAP-028	Capsule Electrical Power Architecture (Independent)	Architecture
100	LSH-CAP-029	Capsule Lighting & Emergency Lighting	Design Description
101	LSH-CAP-030	Capsule Communication & Locator Beacon	Design Description
102	LSH-CAP-031	Capsule Normal-Flight Attachment Load Path	Analysis Report
103	LSH-CAP-032	Capsule Shock Isolation Design	Design Description
104	LSH-CAP-033	Capsule Retention Mechanism Design	Design Description
105	LSH-CAP-034	Capsule Release Interface Definition	ICD
106	LSH-CAP-035	Capsule Structural Substantiation Plan	Verification Plan
107	LSH-CAP-036	Capsule Static Strength Analysis	Analysis Report
108	LSH-CAP-037	Capsule Fatigue & Damage Tolerance Analysis	Analysis Report
109	LSH-CAP-038	Capsule Ultimate Load Cases Definition	Requirements
110	LSH-CAP-039	Capsule Crash Load Cases Definition	Requirements
111	LSH-CAP-040	Capsule Material Allowables & Qualification	Materials
112	LSH-CAP-041	Capsule Manufacturing Definition Dossier	Manufacturing
113	LSH-CAP-042	Capsule Assembly & Integration Procedure	Procedure
114	LSH-CAP-043	Capsule Weight Control Plan	Mass Properties
115	LSH-CAP-044	Capsule Bottom-Up Mass Closure Report	Mass Properties
116	LSH-CAP-045	Capsule Finite Element Model Description	Analysis Report
117	LSH-CAP-046	Capsule Modal & Dynamic Analysis	Analysis Report
118	LSH-CAP-047	Capsule Vibration & Acoustic Fatigue	Analysis Report
119	LSH-CAP-048	Capsule Fire Resistance & Smoke Toxicity	Analysis Report
120	LSH-CAP-049	Capsule Water Landing / Flotation Assessment	Analysis Report
121	LSH-CAP-050	Capsule Soft Soil / Uneven Terrain Landing Analysis	Analysis Report
122	LSH-CAP-051	Capsule Occupant Injury Criteria Compliance	Analysis Report
123	LSH-CAP-052	Capsule Dummy / ATD Test Plan	Test Plan
124	LSH-CAP-053	Capsule Drop Test Plan	Test Plan
125	LSH-CAP-054	Capsule Structural Test Plan	Test Plan
126	LSH-CAP-055	Capsule Environmental Qualification Test Plan	Test Plan
127	LSH-CAP-056	Capsule System Integration Test Plan	Test Plan
128	LSH-CAP-057	Capsule Safety Assessment (FHA / FMEA)	Safety Assessment
129	LSH-CAP-058	Capsule Common-Cause Analysis	Safety Assessment
130	LSH-CAP-059	Capsule Verification Matrix	Traceability
131	LSH-CAP-060	Capsule Evidence Closure Index	Evidence Index
132	LSH-CAP-061	Capsule Maintenance Manual Outline	Maintenance
133	LSH-CAP-062	Capsule Inspection Schedule & NDT Methods	Maintenance
134	LSH-CAP-063	Capsule Handling, Transport & Storage Requirements	Requirements
135	LSH-CAP-064	Capsule Marking, Placards & Emergency Instructions	Design Description
136	LSH-CAP-065	Capsule Human Factors Evaluation Plan	Human Factors
137	LSH-CAP-066	Capsule Crew / Passenger Interface Design	Design Description
138	LSH-CAP-067	Capsule Emergency Equipment Stowage	Design Description
139	LSH-CAP-068	Capsule Pressure / Ventilation Concept (if applicable)	Concept
140	LSH-CAP-069	Capsule Corrosion Prevention & Control	Materials
141	LSH-CAP-070	Capsule Lightning Strike Zoning	Analysis Report
142	LSH-CAP-071	Capsule HIRF & EMC Design Requirements	Requirements
143	LSH-CAP-072	Capsule Software / Firmware Requirements (if any)	Requirements
144	LSH-CAP-073	Capsule Sensor Suite for State Monitoring	Design Description
145	LSH-CAP-074	Capsule Clearance Geometry Definition	Geometry
146	LSH-CAP-075	Capsule Collision Envelope with Separated Parts	Analysis Report
147	LSH-CAP-076	Capsule 6-DOF Dynamic Model Description	Analysis Report
148	LSH-CAP-077	Capsule Aerodynamic Characteristics (Post-Separation)	Analysis Report
149	LSH-CAP-078	Capsule Stability & Control during Descent	Analysis Report
150	LSH-CAP-079	Capsule Ground Impact Energy Management Trade Study	Trade Study
151	LSH-CAP-080	Capsule Multi-Canopy Attachment Points Design	Design Description
152	LSH-CAP-081	Capsule Suspension Load Path Analysis	Analysis Report
153	LSH-CAP-082	Capsule Reefing & Deployment Shock Analysis	Analysis Report
154	LSH-CAP-083	Capsule Post-Landing Stability & Overturn Assessment	Analysis Report
155	LSH-CAP-084	Capsule Rescue Access & External Features	Design Description
156	LSH-CAP-085	Capsule Configuration Control Document	Configuration
157	LSH-CAP-086	Capsule Change Impact Assessment Procedure	Procedure
158	LSH-CAP-087	Capsule Supplier Requirements Document	Requirements
159	LSH-CAP-088	Capsule Quality Assurance Plan	Quality Plan
160	LSH-CAP-089	Capsule First Article Inspection Plan	Inspection Plan

No.	Document ID	Document Title	Document Type
161	LSH-CAP-090	Capsule Non-Conformance & Corrective Action	Procedure
162	LSH-CAP-091	Capsule Reliability Allocation	Reliability
163	LSH-CAP-092	Capsule Maintainability Analysis	Maintainability
164	LSH-CAP-093	Capsule Cost & Weight Trade Register	Trade Study
165	LSH-CAP-094	Capsule Technology Licensing Technical Package	IP Package
166	LSH-CAP-095	Capsule Critical Open Items Register	Risk Register

4. RAPID SEPARATION / JETTISON SYSTEM

Documents 167 – 226 (60 documents)

No.	Document ID	Document Title	Document Type
167	LSH-JET-001	Separation System Requirements Specification	Requirements
168	LSH-JET-002	Separation Architecture Description	Architecture
169	LSH-JET-003	Separation Sequence Definition & Timing Objectives	Requirements
170	LSH-JET-004	Command Validation & Inhibit Logic Design	Design Description
171	LSH-JET-005	Automatic Initiation Path (S1) Design Description	Design Description
172	LSH-JET-006	Backup Automatic Initiation Path (S2) Design	Design Description
173	LSH-JET-007	Manual Crew Initiation Path (S3) Design	Design Description
174	LSH-JET-008	Separation Sensor Suite Requirements	Requirements
175	LSH-JET-009	Separation State Sensing & Clearance Logic	Design Description
176	LSH-JET-010	Release Mechanism Design Description (Primary)	Design Description
177	LSH-JET-011	Release Mechanism Design Description (Secondary)	Design Description
178	LSH-JET-012	Capsule Attachment Release Design	Design Description
179	LSH-JET-013	Retention Loads Analysis — Normal Flight	Analysis Report
180	LSH-JET-014	Release Dynamics Analysis	Analysis Report
181	LSH-JET-015	Failed-Release Case Analysis	Analysis Report
182	LSH-JET-016	Unintended-Release Case Analysis	Analysis Report
183	LSH-JET-017	Asymmetric Separation Dynamics Analysis	Analysis Report
184	LSH-JET-018	Partial Separation Hazard Analysis	Safety Assessment
185	LSH-JET-019	Engine Jettison Feasibility & Trajectory Analysis	Analysis Report
186	LSH-JET-020	Fuel Source Isolation & Release Analysis	Analysis Report
187	LSH-JET-021	Tail Rotor System Separation Analysis	Analysis Report
188	LSH-JET-022	Selected Structure Jettison Analysis	Analysis Report
189	LSH-JET-023	Main Rotor Release Concept Assessment (if pursued)	Trade Study
190	LSH-JET-024	Rotor Inertia & Gyroscopic Effect Analysis	Analysis Report
191	LSH-JET-025	Blade / Hub Trajectory & Clearance Analysis	Analysis Report
192	LSH-JET-026	Component Collision Envelope Analysis	Analysis Report
193	LSH-JET-027	Fuel Isolation Sequencing & Residual Hazard	Analysis Report
194	LSH-JET-028	Pyrotechnic / Actuation Energy Analysis	Analysis Report
195	LSH-JET-029	Release Mechanism Reliability Allocation	Reliability
196	LSH-JET-030	Release Mechanism FMEA / FMECA	Safety Assessment
197	LSH-JET-031	Separation System Common-Cause Analysis	Safety Assessment
198	LSH-JET-032	Separation System ICD (Structure / Power / Logic)	ICD
199	LSH-JET-033	Separation System Power Independence Design	Design Description
200	LSH-JET-034	Separation System EMC & Lightning Protection	Design Description
201	LSH-JET-035	Crew Control Panel & Indication Design	Design Description
202	LSH-JET-036	Inadvertent Actuation Protection Analysis	Safety Assessment
203	LSH-JET-037	Separation System Software Requirements	Requirements
204	LSH-JET-038	Separation System Software Architecture	Architecture
205	LSH-JET-039	Separation Timing Verification Plan	Verification Plan
206	LSH-JET-040	Separation Mechanism Rig Test Plan	Test Plan
207	LSH-JET-041	Separation Sequence Ground Demonstration Plan	Test Plan
208	LSH-JET-042	High-Speed Camera & Instrumentation Plan	Test Plan
209	LSH-JET-043	Separation System Environmental Qualification	Test Plan
210	LSH-JET-044	Separation System Safety Assessment	Safety Assessment
211	LSH-JET-045	Quantitative Unintended Jettison Probability Analysis	Safety Assessment
212	LSH-JET-046	Separation System Verification Matrix	Traceability
213	LSH-JET-047	Separation System Mass Properties	Mass Properties
214	LSH-JET-048	Separation Hardware Manufacturing Definition	Manufacturing
215	LSH-JET-049	Separation System Maintenance & Inspection	Maintenance
216	LSH-JET-050	Separation System Critical Open Items	Risk Register
217	LSH-JET-051	Separation System Trade Studies Register	Trade Study
218	LSH-JET-052	Actuation Authority & Energy Source Selection	Trade Study
219	LSH-JET-053	Separation System Human Factors Assessment	Human Factors
220	LSH-JET-054	Crew Training Requirements for Separation	Requirements
221	LSH-JET-055	Separation System Simulation Model Description	Analysis Report
222	LSH-JET-056	Separation System Monte-Carlo Trajectory Study	Analysis Report

No.	Document ID	Document Title	Document Type
223	LSH-JET-057	Separation System Interface with Parachute Logic	ICD
224	LSH-JET-058	Separation System Configuration Control	Configuration
225	LSH-JET-059	Separation System Supplier Requirements	Requirements
226	LSH-JET-060	Separation System Evidence Closure Index	Evidence Index

5. PARACHUTE RECOVERY SYSTEM

Documents 227 – 261 (35 documents)

No.	Document ID	Document Title	Document Type
227	LSH-PAR-001	Parachute System Requirements Specification	Requirements
228	LSH-PAR-002	Parachute Architecture & Canopy Count Trade Study	Trade Study
229	LSH-PAR-003	Parachute Sizing Analysis (Mass / Descent Rate)	Analysis Report
230	LSH-PAR-004	Deployment Envelope Definition	Requirements
231	LSH-PAR-005	Deployment Sequence & Reefing Design	Design Description
232	LSH-PAR-006	Canopy Design Description	Design Description
233	LSH-PAR-007	Suspension Line & Riser Design	Design Description
234	LSH-PAR-008	Deployment Bag & Packing Design	Design Description
235	LSH-PAR-009	Pilot Chute / Extraction System Design	Design Description
236	LSH-PAR-010	Inflation Load Analysis	Analysis Report
237	LSH-PAR-011	Opening Shock & Peak Load Analysis	Analysis Report
238	LSH-PAR-012	Asymmetric Inflation & Oscillation Analysis	Analysis Report
239	LSH-PAR-013	Single-Canopy Failure Case Analysis	Analysis Report
240	LSH-PAR-014	Line / Canopy Damage Case Analysis	Analysis Report
241	LSH-PAR-015	Parachute Reliability Allocation & Prediction	Reliability
242	LSH-PAR-016	Parachute System FMEA	Safety Assessment
243	LSH-PAR-017	Parachute Clearance Logic Interface	ICD
244	LSH-PAR-018	Prohibited Deployment Conditions Definition	Requirements
245	LSH-PAR-019	Parachute Material Specification & Qualification	Materials
246	LSH-PAR-020	Parachute Environmental Qualification Plan	Test Plan
247	LSH-PAR-021	Parachute Component Test Plan	Test Plan
248	LSH-PAR-022	Parachute Deployment Test Plan	Test Plan
249	LSH-PAR-023	Parachute System-Level Demonstration Plan	Test Plan
250	LSH-PAR-024	Parachute Stability & Descent Rate Verification	Verification Plan
251	LSH-PAR-025	Parachute Mass Properties	Mass Properties
252	LSH-PAR-026	Parachute Attachment to Capsule Design	Design Description
253	LSH-PAR-027	Parachute Load Path into Capsule Structure	Analysis Report
254	LSH-PAR-028	Parachute Packing, Storage & Life Limits	Maintenance
255	LSH-PAR-029	Parachute Inspection & Repack Procedure	Procedure
256	LSH-PAR-030	Parachute System Safety Assessment	Safety Assessment
257	LSH-PAR-031	Parachute Verification Matrix	Traceability
258	LSH-PAR-032	Parachute Supplier Requirements Document	Requirements
259	LSH-PAR-033	Parachute Critical Open Items Register	Risk Register
260	LSH-PAR-034	Parachute Technology Licensing Package	IP Package
261	LSH-PAR-035	Parachute Evidence Closure Index	Evidence Index

6. MAIN AIRFRAME & HELICOPTER SYSTEMS

Documents 262 – 331 (70 documents)

No.	Document ID	Document Title	Document Type
262	LSH-AF-001	Airframe System Requirements Specification	Requirements
263	LSH-AF-002	Airframe Architecture Description	Architecture
264	LSH-AF-003	Fuselage Design Description	Design Description
265	LSH-AF-004	Fuselage Structural Analysis	Analysis Report
266	LSH-AF-005	Landing Gear / Skid Design Description	Design Description
267	LSH-AF-006	Landing Gear Loads Analysis	Analysis Report
268	LSH-AF-007	Airframe Mass Properties Report	Mass Properties
269	LSH-AF-008	Airframe CG Envelope	Mass Properties
270	LSH-AF-009	Airframe Interface with Capsule ICD	ICD
271	LSH-AF-010	Airframe / Capsule Attachment Structure Design	Design Description
272	LSH-AF-011	Airframe Aerodynamic Design Description	Design Description
273	LSH-AF-012	Airframe Drag Build-Up Analysis	Analysis Report
274	LSH-AF-013	Airframe Vibration & Dynamic Response	Analysis Report
275	LSH-AF-014	Airframe Fatigue Analysis	Analysis Report
276	LSH-AF-015	Airframe Damage Tolerance Assessment	Analysis Report
277	LSH-AF-016	Airframe Material Selection & Allowables	Materials
278	LSH-AF-017	Airframe Corrosion Prevention Plan	Materials
279	LSH-AF-018	Airframe Manufacturing Definition	Manufacturing

No.	Document ID	Document Title	Document Type
280	LSH-AF-019	Airframe Assembly Sequence	Procedure
281	LSH-AF-020	Cockpit Arrangement & Human Factors	Human Factors
282	LSH-AF-021	Crew Station Design Description	Design Description
283	LSH-AF-022	Flight Control System Architecture	Architecture
284	LSH-AF-023	Flight Control Loads Analysis	Analysis Report
285	LSH-AF-024	Hydraulic System Design Description	Design Description
286	LSH-AF-025	Hydraulic System Analysis	Analysis Report
287	LSH-AF-026	Electrical System Architecture	Architecture
288	LSH-AF-027	Electrical Load Analysis	Analysis Report
289	LSH-AF-028	Electrical Power Generation & Distribution	Design Description
290	LSH-AF-029	Emergency Electrical Power Design	Design Description
291	LSH-AF-030	Avionics Architecture Description	Architecture
292	LSH-AF-031	Avionics ICD Master	ICD
293	LSH-AF-032	Flight Management & Navigation System	Design Description
294	LSH-AF-033	Communication System Design	Design Description
295	LSH-AF-034	Surveillance & Transponder System	Design Description
296	LSH-AF-035	Health & Usage Monitoring System (HUMS)	Design Description
297	LSH-AF-036	Cockpit Display System Design	Design Description
298	LSH-AF-037	Warning & Caution System Design	Design Description
299	LSH-AF-038	Fuel System Design Description	Design Description
300	LSH-AF-039	Fuel System Analysis & Tank Geometry	Analysis Report
301	LSH-AF-040	Fuel Isolation & Fire Protection Design	Design Description
302	LSH-AF-041	Fuel System Safety Assessment	Safety Assessment
303	LSH-AF-042	Environmental Control System Design	Design Description
304	LSH-AF-043	Ice Protection System Design	Design Description
305	LSH-AF-044	Lighting System Design (External / Internal)	Design Description
306	LSH-AF-045	Airframe Fire Detection & Suppression	Design Description
307	LSH-AF-046	Airframe Structural Test Plan	Test Plan
308	LSH-AF-047	Airframe Ground Vibration Test Plan	Test Plan
309	LSH-AF-048	Airframe Systems Integration Test Plan	Test Plan
310	LSH-AF-049	Airframe Safety Assessment (FHA / FMEA)	Safety Assessment
311	LSH-AF-050	Airframe Verification Matrix	Traceability
312	LSH-AF-051	Airframe Weight Control Plan	Mass Properties
313	LSH-AF-052	Airframe Maintenance Manual Outline	Maintenance
314	LSH-AF-053	Airframe Inspection Program	Maintenance
315	LSH-AF-054	Airframe Configuration Control Document	Configuration
316	LSH-AF-055	Airframe Supplier Requirements	Requirements
317	LSH-AF-056	Airframe Quality Assurance Plan	Quality Plan
318	LSH-AF-057	Airframe Critical Open Items	Risk Register
319	LSH-AF-058	Cabin Interior Design Description	Design Description
320	LSH-AF-059	Cabin Safety & Emergency Equipment	Design Description
321	LSH-AF-060	Airframe Lightning Zoning & Protection	Analysis Report
322	LSH-AF-061	Airframe HIRF Protection Design	Design Description
323	LSH-AF-062	Airframe Bird Strike Assessment	Analysis Report
324	LSH-AF-063	Airframe Crashworthiness Integration with Capsule	Analysis Report
325	LSH-AF-064	Airframe / Capsule Combined Mass Model	Mass Properties
326	LSH-AF-065	Airframe Geometry Control Document	Geometry
327	LSH-AF-066	Airframe Digital Mock-Up Definition	Design Description
328	LSH-AF-067	Airframe Tolerance & Interface Management	Configuration
329	LSH-AF-068	Airframe Environmental Qualification Plan	Test Plan
330	LSH-AF-069	Airframe Evidence Closure Index	Evidence Index
331	LSH-AF-070	Airframe Technology Licensing Package	IP Package

7. PROPULSION, ROTOR & TRANSMISSION

Documents 332 – 381 (50 documents)

No.	Document ID	Document Title	Document Type
332	LSH-PRO-001	Propulsion System Requirements Specification	Requirements
333	LSH-PRO-002	Propulsion Architecture Description	Architecture
334	LSH-PRO-003	Engine Selection & Study Range Definition	Trade Study
335	LSH-PRO-004	Power Required vs Power Available Analysis	Analysis Report
336	LSH-PRO-005	Hover Performance Analysis	Analysis Report
337	LSH-PRO-006	Climb Performance Analysis	Analysis Report
338	LSH-PRO-007	Cruise Performance Analysis	Analysis Report
339	LSH-PRO-008	OEI Performance Analysis	Analysis Report
340	LSH-PRO-009	Autorotation Performance Analysis	Analysis Report
341	LSH-PRO-010	Hot / High Performance Analysis	Analysis Report

No.	Document ID	Document Title	Document Type
342	LSH-PRO-011	Transmission Design Description	Design Description
343	LSH-PRO-012	Transmission Loads & Torque Analysis	Analysis Report
344	LSH-PRO-013	Transmission Thermal Analysis	Analysis Report
345	LSH-PRO-014	Main Rotor Design Description	Design Description
346	LSH-PRO-015	Main Rotor Aerodynamic Performance Model	Analysis Report
347	LSH-PRO-016	Main Rotor Structural Analysis	Analysis Report
348	LSH-PRO-017	Main Rotor Hub & Controls Design	Design Description
349	LSH-PRO-018	Main Rotor Loads Analysis	Analysis Report
350	LSH-PRO-019	Main Rotor Vibration Analysis	Analysis Report
351	LSH-PRO-020	Tail Rotor Design Description	Design Description
352	LSH-PRO-021	Tail Rotor Performance & Loads Analysis	Analysis Report
353	LSH-PRO-022	Drive Shafting Design Description	Design Description
354	LSH-PRO-023	Engine Mount Design & Loads	Design Description
355	LSH-PRO-024	Engine Installation & Cooling Analysis	Analysis Report
356	LSH-PRO-025	Engine Control System Interface	ICD
357	LSH-PRO-026	Propulsion System Mass Properties	Mass Properties
358	LSH-PRO-027	Propulsion System Safety Assessment	Safety Assessment
359	LSH-PRO-028	Propulsion FMEA / FMECA	Safety Assessment
360	LSH-PRO-029	Propulsion Common-Cause Analysis	Safety Assessment
361	LSH-PRO-030	Propulsion / Separation Interface Analysis	Analysis Report
362	LSH-PRO-031	Engine Jettison Interface Definition (if applicable)	ICD
363	LSH-PRO-032	Rotor Brake & Rotor Stopping Analysis	Analysis Report
364	LSH-PRO-033	Propulsion System Test Plan	Test Plan
365	LSH-PRO-034	Engine Ground Run & Acceptance Test Plan	Test Plan
366	LSH-PRO-035	Transmission Bench Test Plan	Test Plan
367	LSH-PRO-036	Rotor System Ground Test Plan	Test Plan
368	LSH-PRO-037	Propulsion System Verification Matrix	Traceability
369	LSH-PRO-038	Propulsion System ICD Master	ICD
370	LSH-PRO-039	Propulsion System Maintenance Requirements	Maintenance
371	LSH-PRO-040	Propulsion System Configuration Control	Configuration
372	LSH-PRO-041	Propulsion Critical Open Items	Risk Register
373	LSH-PRO-042	Figure of Merit & Rotor Efficiency Study	Analysis Report
374	LSH-PRO-043	Rotor Noise Assessment	Analysis Report
375	LSH-PRO-044	Propulsion System Environmental Qualification	Test Plan
376	LSH-PRO-045	Propulsion System Reliability Allocation	Reliability
377	LSH-PRO-046	Propulsion System Supplier Requirements	Requirements
378	LSH-PRO-047	Propulsion System Evidence Closure Index	Evidence Index
379	LSH-PRO-048	Power Assurance & Trend Monitoring	Design Description
380	LSH-PRO-049	Propulsion System Trade Studies Register	Trade Study
381	LSH-PRO-050	Propulsion Technology Licensing Package	IP Package

8. AVIONICS, SOFTWARE, CONTROL LOGIC & HUMAN FACTORS

Documents 382 – 421 (40 documents)

No.	Document ID	Document Title	Document Type
382	LSH-AV-001	Avionics System Requirements Specification	Requirements
383	LSH-AV-002	Avionics Architecture Description	Architecture
384	LSH-AV-003	Emergency Logic Requirements Specification	Requirements
385	LSH-AV-004	Automatic Initiation Logic Design Description	Design Description
386	LSH-AV-005	Manual Initiation Logic & Crew Interface	Design Description
387	LSH-AV-006	Sensor Validity & Voting Logic Design	Design Description
388	LSH-AV-007	Inhibit, Arming & Timing Logic Design	Design Description
389	LSH-AV-008	Fault Handling & Failure Annunciation Design	Design Description
390	LSH-AV-009	Power Interruption & Degraded Mode Logic	Design Description
391	LSH-AV-010	Software Requirements Specification — Emergency System	Requirements
392	LSH-AV-011	Software Architecture Description — Emergency System	Architecture
393	LSH-AV-012	Software Design Description — Emergency System	Design Description
394	LSH-AV-013	Software Verification & Validation Plan	Verification Plan
395	LSH-AV-014	Software Criticality Assessment (DAL)	Safety Assessment
396	LSH-AV-015	Complex Electronic Hardware Plan	Assurance Plan
397	LSH-AV-016	Avionics EMC & HIRF Design	Design Description
398	LSH-AV-017	Avionics Power Supply Design	Design Description
399	LSH-AV-018	Data Bus & Communication Architecture	Architecture
400	LSH-AV-019	Cockpit Human Factors Analysis	Human Factors
401	LSH-AV-020	Crew Workload Assessment — Emergency Scenarios	Human Factors
402	LSH-AV-021	Inadvertent Actuation Human Factors Assessment	Human Factors
403	LSH-AV-022	Crew Training Needs Analysis	Human Factors

No.	Document ID	Document Title	Document Type
404	LSH-AV-023	Cockpit Display & Alert Design Guidelines	Design Description
405	LSH-AV-024	Avionics System Safety Assessment	Safety Assessment
406	LSH-AV-025	Avionics FMEA / FMECA	Safety Assessment
407	LSH-AV-026	Avionics Common-Cause Analysis	Safety Assessment
408	LSH-AV-027	Avionics ICD Master	ICD
409	LSH-AV-028	Avionics Integration Test Plan	Test Plan
410	LSH-AV-029	Software Test Plan	Test Plan
411	LSH-AV-030	Hardware / Software Integration Test Plan	Test Plan
412	LSH-AV-031	Avionics Environmental Qualification Plan	Test Plan
413	LSH-AV-032	Avionics Verification Matrix	Traceability
414	LSH-AV-033	Avionics Configuration Control	Configuration
415	LSH-AV-034	Avionics Supplier Requirements	Requirements
416	LSH-AV-035	Avionics Critical Open Items	Risk Register
417	LSH-AV-036	Flight Control Computer Requirements (if applicable)	Requirements
418	LSH-AV-037	Autopilot / AFCS Interface with Emergency Logic	ICD
419	LSH-AV-038	Data Recording & Post-Event Analysis Requirements	Requirements
420	LSH-AV-039	Avionics Evidence Closure Index	Evidence Index
421	LSH-AV-040	Avionics Technology Licensing Package	IP Package

9. PERFORMANCE, LOADS, DYNAMICS & MISSION ANALYSIS

Documents 422 – 461 (40 documents)

No.	Document ID	Document Title	Document Type
422	LSH-PER-001	Aircraft Performance Requirements Specification	Requirements
423	LSH-PER-002	Mission Analysis & Mission Profiles	Analysis Report
424	LSH-PER-003	Range & Endurance Analysis	Analysis Report
425	LSH-PER-004	Payload / Range Diagram	Analysis Report
426	LSH-PER-005	Hover Ceiling Analysis	Analysis Report
427	LSH-PER-006	Forward Flight Envelope Definition	Requirements
428	LSH-PER-007	Structural Loads Envelope Definition	Requirements
429	LSH-PER-008	Flight Loads Analysis — Maneuver	Analysis Report
430	LSH-PER-009	Flight Loads Analysis — Gust	Analysis Report
431	LSH-PER-010	Ground Loads Analysis	Analysis Report
432	LSH-PER-011	Landing Loads Analysis (Normal & Emergency)	Analysis Report
433	LSH-PER-012	Separation Loads Analysis	Analysis Report
434	LSH-PER-013	Parachute Opening Loads Analysis	Analysis Report
435	LSH-PER-014	Combined Aircraft + Capsule Loads Model	Analysis Report
436	LSH-PER-015	6-DOF Aircraft Dynamic Model Description	Analysis Report
437	LSH-PER-016	6-DOF Capsule Post-Separation Dynamic Model	Analysis Report
438	LSH-PER-017	Aeroelastic & Whirl Flutter Assessment	Analysis Report
439	LSH-PER-018	Ground Resonance Assessment	Analysis Report
440	LSH-PER-019	Rotor / Airframe Coupling Analysis	Analysis Report
441	LSH-PER-020	Noise Certification Assessment	Analysis Report
442	LSH-PER-021	Performance Model Validation Plan	Verification Plan
443	LSH-PER-022	Loads Model Validation Plan	Verification Plan
444	LSH-PER-023	Flight Test Instrumentation Requirements	Requirements
445	LSH-PER-024	Flight Test Plan — Performance	Test Plan
446	LSH-PER-025	Flight Test Plan — Handling Qualities	Test Plan
447	LSH-PER-026	Flight Test Plan — Loads	Test Plan
448	LSH-PER-027	Flight Test Plan — Systems	Test Plan
449	LSH-PER-028	Flight Test Safety Assessment	Safety Assessment
450	LSH-PER-029	Performance & Loads Verification Matrix	Traceability
451	LSH-PER-030	Performance Critical Open Items	Risk Register
452	LSH-PER-031	Weight & Balance Manual Outline	Mass Properties
453	LSH-PER-032	CG Travel & Loading Schedule	Mass Properties
454	LSH-PER-033	Inertia Measurement & Validation Plan	Verification Plan
455	LSH-PER-034	Environmental Envelope Definition	Requirements
456	LSH-PER-035	Operating Limitations Definition	Requirements
457	LSH-PER-036	Emergency Operating Procedures Outline	Procedure
458	LSH-PER-037	Performance Evidence Closure Index	Evidence Index
459	LSH-PER-038	Handling Qualities Requirements	Requirements
460	LSH-PER-039	Stability & Control Analysis	Analysis Report
461	LSH-PER-040	Mission Suitability Assessment	Assessment

10. VERIFICATION, VALIDATION, TEST & EVIDENCE

Documents 462 – 501 (40 documents)

No.	Document ID	Document Title	Document Type
462	LSH-VV-001	Master Verification & Validation Plan	Verification Plan
463	LSH-VV-002	Verification Requirements Allocation Matrix	Traceability
464	LSH-VV-003	Test & Evaluation Master Plan (TEMP)	Verification Plan
465	LSH-VV-004	Ground Test Plan — Integrated	Test Plan
466	LSH-VV-005	Structural Test Plan — Full Scale	Test Plan
467	LSH-VV-006	Systems Integration Test Plan	Test Plan
468	LSH-VV-007	Electromagnetic Compatibility Test Plan	Test Plan
469	LSH-VV-008	Environmental Qualification Test Plan — System	Test Plan
470	LSH-VV-009	Lightning Test Plan	Test Plan
471	LSH-VV-010	HIRF Test Plan	Test Plan
472	LSH-VV-011	Software Verification Plan	Verification Plan
473	LSH-VV-012	Hardware / Software Integration Test Plan	Test Plan
474	LSH-VV-013	Separation System Demonstration Plan	Test Plan
475	LSH-VV-014	Capsule Drop & Impact Test Plan	Test Plan
476	LSH-VV-015	Parachute Deployment Test Plan	Test Plan
477	LSH-VV-016	Occupant Protection Test Plan	Test Plan
478	LSH-VV-017	Flight Test Plan — Overall	Test Plan
479	LSH-VV-018	Flight Test Instrumentation & Data Plan	Test Plan
480	LSH-VV-019	Flight Test Safety Plan	Safety Assessment
481	LSH-VV-020	Test Article Configuration Control	Configuration
482	LSH-VV-021	Test Readiness Review Procedure	Procedure
483	LSH-VV-022	Anomaly Reporting & Corrective Action Procedure	Procedure
484	LSH-VV-023	Evidence Collection & Archiving Procedure	Procedure
485	LSH-VV-024	Verification Gate Closure Reports (VG-01 to VG-12)	Evidence Index
486	LSH-VV-025	Requirements Verification Summary Report	Evidence Index
487	LSH-VV-026	Safety Evidence Summary Report	Evidence Index
488	LSH-VV-027	Performance Evidence Summary Report	Evidence Index
489	LSH-VV-028	Structural Evidence Summary Report	Evidence Index
490	LSH-VV-029	Separation System Evidence Summary	Evidence Index
491	LSH-VV-030	Parachute System Evidence Summary	Evidence Index
492	LSH-VV-031	Capsule System Evidence Summary	Evidence Index
493	LSH-VV-032	Independent Verification & Validation Plan	Assurance Plan
494	LSH-VV-033	Conformity Inspection Plan	Inspection Plan
495	LSH-VV-034	First Article Inspection Plan — Critical Items	Inspection Plan
496	LSH-VV-035	Test Facility & Equipment Requirements	Requirements
497	LSH-VV-036	Instrumentation Calibration & Uncertainty	Procedure
498	LSH-VV-037	Data Analysis & Reporting Standards	Procedure
499	LSH-VV-038	Verification Critical Open Items	Risk Register
500	LSH-VV-039	V&V; Evidence Closure Index Master	Evidence Index
501	LSH-VV-040	Certification Flight Test Plan Outline	Test Plan

11. MANUFACTURING, QUALITY, CONFIGURATION & IP

Documents 502 – 531 (30 documents)

No.	Document ID	Document Title	Document Type
502	LSH-MFG-001	Manufacturing Plan — System Level	Manufacturing
503	LSH-MFG-002	Build-to-Print & Model-Based Definition Strategy	Manufacturing
504	LSH-MFG-003	Tooling & Fixture Requirements	Manufacturing
505	LSH-MFG-004	Special Process Specifications List	Manufacturing
506	LSH-MFG-005	Composite Manufacturing Process Specification	Manufacturing
507	LSH-MFG-006	Metallic Parts Manufacturing Specification	Manufacturing
508	LSH-MFG-007	Assembly Process Specification — Capsule	Manufacturing
509	LSH-MFG-008	Assembly Process Specification — Airframe	Manufacturing
510	LSH-MFG-009	Quality Management System Plan	Quality Plan
511	LSH-MFG-010	Inspection & Test Plan Master	Inspection Plan
512	LSH-MFG-011	Non-Conformance Control Procedure	Procedure
513	LSH-MFG-012	Supplier Quality Requirements	Requirements
514	LSH-MFG-013	First Article Inspection Procedure	Procedure
515	LSH-MFG-014	Configuration Management Plan (Detailed)	Management Plan
516	LSH-MFG-015	Change Control Board Charter	Management Plan
517	LSH-MFG-016	Product Data Management Procedure	Procedure
518	LSH-MFG-017	As-Built Configuration Recording	Configuration
519	LSH-MFG-018	IP Asset Identification & Protection Plan	IP Package
520	LSH-MFG-019	Patent / Trade Secret Register	IP Package
521	LSH-MFG-020	Know-How Documentation Package	IP Package
522	LSH-MFG-021	Software & Data Rights Definition	IP Package
523	LSH-MFG-022	Technology Licensing Technical Data Package	IP Package

No.	Document ID	Document Title	Document Type
524	LSH-MFG-023	Export Control & Classification Assessment	IP Package
525	LSH-MFG-024	Obsolescence Management Plan	Management Plan
526	LSH-MFG-025	Packaging, Handling, Storage & Transportation	Procedure
527	LSH-MFG-026	Production Readiness Review Plan	Management Plan
528	LSH-MFG-027	Manufacturing Critical Open Items	Risk Register
529	LSH-MFG-028	Cost Estimating Model & Basis of Estimate	Management Plan
530	LSH-MFG-029	Make-or-Buy Decision Register	Trade Study
531	LSH-MFG-030	Manufacturing Evidence & Record Index	Evidence Index

END OF LSH 00 ENGINEERING DOCUMENT TREE

Sequential numbering 001–531 is provided for overall tracking. Logical Document IDs remain the configuration-controlled identifiers. No document claims certification approval or demonstrated performance unless objective evidence is formally accepted.

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